Hasibur Rahman

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PROFESSIONAL SUMMARY

Results-driven AI Engineer with hands-on leadership experience guiding cross-functional teams in building production-grade AI systems. **Proven track record leading 3–4 engineer teams** to deliver end-to-end voice AI platforms, fine-tune ASR models for **77% faster inference**, architect secure RAG pipelines, and **implement agentic workflows tested at 1M-step scale**. Expert in NLP, ASR/TTS, LLM orchestration, MLOps, and multilingual AI solutions.

EXPERIENCE

VivaSoft Ltd
June 2024 – Present

Software Engineer
Dhaka, Bangladesh

- **Currently leading Kothon Analytics**, a production **Voice AI + Business Intelligence platform** that processes call-center conversations into actionable business insights, profiling, issue detection, escalation signals, and owner-level analytics processing over **125+ hours of calls daily**
- **Led 4-engineer team architecting 6 scalable Golang microservices** with strong concurrency, fault tolerance and observability
- Fine-tuned transformer-based Bangla ASR model; **achieved 38% faster inference** via custom tokenizer and efficient Go processing
- Designed clean REST APIs; deployed containerised services on GCP (Cloud Run, Redis, GCS) using Terraform and Docker
- Engineered production RAG pipeline achieving **100% Faithfulness**, **100% Conversation Completeness**, **93% Knowledge Retention**, and **100% GEval security score** across multi-turn call-center conversation analysis
- Achieved **7/8 pass rate** across evaluation dimensions including trajectory correctness, safety, faithfulness, and multi-turn coherence using **DeepEval-based automated benchmarking**
- Optimized analytics and agentic orchestration ensuring **Star Schema** for faster reporting
- Integrated **MAKER** mechanism to ensure reliable agentic decision **tested on 1M step task**
- Engineered a production-grade **MCP (Model Context Protocol) server** with **header-based authentication**, interoperable with **OpenCode, Cline**, and other MCP-compliant AI clients — turning internal analytics into agent-callable tools
- Architected an event-sourced **Fact/Event Store** backed by an asynchronous **RabbitMQ** pipeline with dead-letter recovery, built for reliable ingestion at scale
- Designed **Redis-backed** referential-integrity safeguards protecting critical AI data pipelines from silent corruption
- **Tech Stack:** Golang, React, FastAPI, Whisper, Langchain, Gemini, Redis, RabbitMQ, MCP (header-based auth), GCP, Docker, Terraform, Prometheus, Grafana, Microservices

Chaldal (YC-backed e-commerce, 1.2M users)
March 2023 – June 2024

Software Engineer
Dhaka, Bangladesh

- Built fraud prevention logic **reducing loss by 40tk/order** using efficient, **sargable database queries** and clean backend code
- Developed address change flow **used by 1M+ users**; **improved saving efficiency by 72%**
- **Tech Stack:** ASP.NET Core, TypeScript, React, React Native, Azure, Entity Framework, Hangfire

PROJECTS

AI Voice Calendar Scheduler
2025–2026

- Self-hosted voice agent for real-time scheduling: **WebRTC conversations, Google Calendar** conflict checks, meeting creation, confirmation emails

- Privacy-focused with fully self-hosted STT/TTS pipeline; exposes tools via **MCP server/client**
- Containerized deployment; follows Clean Architecture
- **Tech Stack:** Python, LiveKit (WebRTC), Google Gemini, Faster-Whisper, Piper TTS, Google Calendar API, Docker, MCP

WhatsApp-Integrated Multi-Agent Business Intelligence 2026

- Architected a modular Multi-Agent System (MAS) with **4 specialized agents** for autonomous strategic task delegation via WhatsApp.
- Developed an **Investigative Discovery Loop cross-referencing 10+ API endpoints** to transform raw data into context-aware RAG insights.
- Automated executive reporting from 1000s of data points using a "No Excuses" protocol for comprehensive narrative synthesis.
- **Tech Stack:** FastAPI (Python), genai, WhatsApp Business API, Docker, Pydantic, PostgreSQL

Intelligent Legal Multi-Agent RAG 2026-Present

- Architected a modular **Multi-Agent System (MAS) with 4 specialized agents** (Semantic, Entity, Temporal, and Fusion) for autonomous parallelized **Chain-of-Retrieval** reasoning.
- Implemented Clean Architecture using FastAPI and Pydantic, achieving 100% architectural decoupling and seamless integration with Gemini
- Developed an **Investigative Discovery Loop** cross-referencing **Qdrant vector embeddings** with **Neo4j** graph entities to transform raw case data into context-aware RAG insights.
- Engineered a **Query Decomposer** and **Rewriter** layer to autonomously decompose complex legal mandates into verifiable sub-tasks, increasing retrieval precision.
- **Tech Stack:** FastAPI (Python), genai sdk, Qdrant (Vector DB), Neo4j (Graph DB), Redis (Vector Cache), Docker, Pydantic, RQ

Technical Skills

Languages:	Python, TypeScript, Go, C#
AI & ML Frameworks:	TensorFlow, PyTorch, Hugging Face Transformers, FastAPI, LiveKit, DeepEval
Libraries:	NumPy, Pandas, Scikit-learn, Matplotlib, LangChain, FairSeq, Nemo
Databases:	Weaviate, Redis Vector DB, MySQL, PostgreSQL
Messaging & Caching:	RabbitMQ (DLX/DLQ patterns), Redis (write-through caching)
Observability & Tools:	Grafana, Prometheus, Azure AppInsights
MLOps & Infra:	Docker, Terraform, AWS, Google Cloud Platform (GCP)
Specialized Domains:	Automatic Speech Recognition (ASR), Text-to-Speech (TTS), Retrieval-Augmented Generation (RAG), Function Calling, Natural Language Processing (NLP), LLM Orchestration, Model Context Protocol (MCP)
Open-Source Contribution:	Bengali ASR Model

AWARDS & ACHIEVEMENTS

2025	**Top 6% (42/1244)**	HackerEarth Machine Learning challenge: World Water Day
2025	**Top 34.5%**	Predicting Road Accident Risk
2021	**Top 10% (185/1747),**	2021 ICPC Asia Dhaka Regional Online Preliminary
2021	**Divisional Champion,**	ICPC Asia Dhaka Regional Contest 2021
2022	**Top 12% (206/1648)**	2022 ICPC Asia Dhaka Regional Online Preliminary

EDUCATION

Jatiya Kabi Kazi Nazrul Islam University March 2018 – February 2023
BSc in Computer Science & Engineering Trishal, Mymensingh, Bangladesh